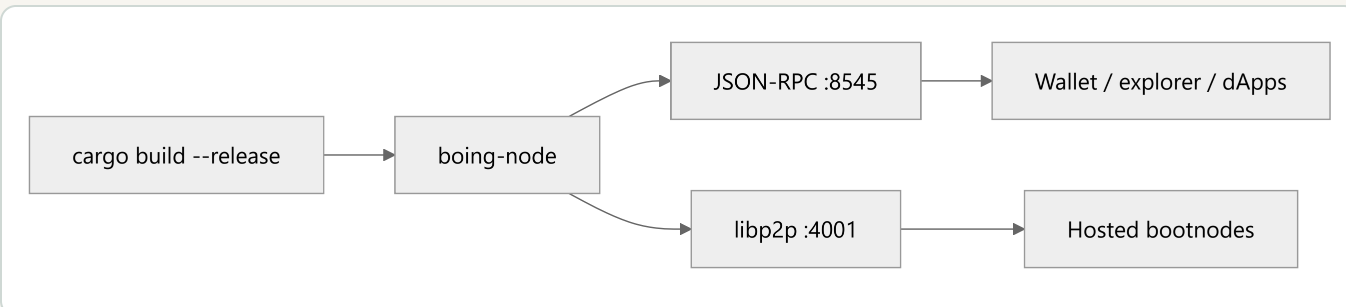
Boing Network — Operational Runbook

> 👋 **Everyday users:** you probably want the faucet and VibeMiner, not this file.
> 🛠️ **Developers:** local node is `cargo run -p boing-node` on port **8545**.
> 🛰️ **Operators:** this is the runbook — setup, RPC, CLI, monitoring, incidents, testnet ops.

Purpose: Operations guide for running and maintaining Boing Network nodes.
References: BOING-NETWORK-ESSENTIALS.md, BUILD-ROADMAP.md, README.md, RPC-API-SPEC.md, READINESS.md, TESTNET-RPC-INFRA.md



Network essentials (six pillars)

The network prioritizes, in order: **1. Security** → **2. Scalability** → **3. Decentralization** → **4. Authenticity** → **5. Transparency** → **6. True quality assurance**. For the full description of each pillar and design philosophy, see BOING-NETWORK-ESSENTIALS.md.

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1. Node Setup

Prerequisites

- **Rust:** 1.70+ (`rustup` recommended)
- **OS:** Linux, macOS, or Windows (WSL recommended on Windows)

Build

```
cargo build --release
```

Directory Layout

Path	Description
`target/release/boing-node`	Node binary
`target/release/boing`	CLI binary
`~/.boing/` or `./data/`	Data directory (when using `--data-dir`)

2. Running a Node

Full Node (non-validator)

```
cargo run -p boing-node
```

Defaults: RPC on `http://127.0.0.1:8545`.

Validator Node

```
cargo run -p boing-node -- --validator --rpc-port 8545
```

Produces blocks when there are pending transactions.

With Data Directory

```
cargo run -p boing-node -- --data-dir ./boing-data --rpc-port 8545
```

Persists chain, QA policy (`qa_registry.json` / `qa_pool_config.json`), and native fee policy (`network_fee_config.json`). Override at start with `--qa-registry`, `--qa-pool-config`, and `--network-fee-config`. Canonical shapes: `docs/config/qa_pool_config.canonical.json`, `docs/config/network_fee_config.canonical.json`. Default fee split is **100% to `PROTOCOL_TREASURY`**; mining emission still credits the round proposer.

Mempool: pending transactions per sender

Each account may have only **N** distinct pending nonces in the mempool at once (default **N = 16**, aligned with `RateLimitConfig::default_mainnet`). The process logs `Mempool: max pending txs per sender = …` at startup. Operators can raise the cap (e.g. busy testnet RPC) with:

```
cargo run -p boing-node -- --validator --rpc-port 8545 --pending-txs-per-sender 64
```

Nonce **replacements** for the same nonce do not count toward the cap.

Dev rate-limit profile (local / busy testnet)

For **relaxed** JSON-RPC rate limits and a **64** pending-tx cap per sender (see `RateLimitConfig::default_devnet`):

```
cargo run -p boing-node -- --validator --rpc-port 8545 --dev-rate-limits
```

Same effect without the flag (e.g. Docker / systemd):

```
export BOING_RATE_PROFILE=dev
cargo run -p boing-node -- --validator --rpc-port 8545
```

Use **`BOING_RATE_PROFILE=mainnet`** to force the strict profile even if **`--dev-rate-limits`** is mistakenly present on a public RPC host.

You can still override only the mempool cap: `--dev-rate-limits --pending-txs-per-sender 128`. **Do not** use the dev profile on public mainnet RPC endpoints.

JSON-RPC surface (stale binary or proxy)

If **`boing_chainHeight`** works but other **`boing_*`** calls return **`-32601`** **Method not found**, the process on the RPC port is almost certainly an **older `boing-node` build** or a **proxy that drops unknown methods**. Rebuild from the current repository (`cargo build -p boing-node --release`) and restart. To see which read-only methods the URL exposes, run **`npm run build`** then **`npm run probe-rpc`** in **`boing-sdk`**, or **`npm run probe-rpc`** from the repository root (prints a **`diagnosis`** when methods are missing). **No SDK build:** from the repo root run **`npm run rpc-endpoint-check`** for a raw JSON-RPC table. **Windows — who owns port 8545:** `netstat -ano | findstr :8545`, then `tasklist /FI "PID eq <pid>" /V` (ensure the image is the **`boing-node`** you just built, not an old `cargo install` path). The tutorial package also exposes **`npm run probe-rpc`**. See RPC-API-SPEC.md § Diagnosing `-32601`.

Public RPC — chain id for dApps: Set environment **`BOING_CHAIN_ID=6913`** and **`BOING_CHAIN_NAME=Boing Testnet`** on the **`boing-node`** process so **`boing_getNetworkInfo`** returns them (matches wallet **`boing_chainId`** **`0x1b01`**). Copy-paste template: `tools/boing-node-public-testnet.env.example`. **systemd / Docker / Kubernetes:** `tools/boing-node-public-testnet.systemd.example`, `tools/boing-node-public-testnet.docker-compose.yml`, `tools/boing-node-public-testnet.kubernetes.example.yaml` — see `tools/README.md`. Wallets can still use a fixed chain id from app config when the field is **`null`**.

Canonical native DEX addresses (optional, recommended for public RPC): Set `BOING_CANONICAL_NATIVE_CP_POOL`, `BOING_CANONICAL_NATIVE_DEX_FACTORY`, and (when deployed) `BOING_CANONICAL_NATIVE_DEX_MULTIHOP_SWAP_ROUTER`, `BOING_CANONICAL_NATIVE_DEX_LEDGER_ROUTER_V2`, `BOING_CANONICAL_NATIVE_DEX_LEDGER_ROUTER_V3`, `BOING_CANONICAL_NATIVE_AMM_LP_VAULT`, `BOING_CANONICAL_NATIVE_LP_SHARE_TOKEN` to **32-byte `AccountId` hex** (64 hex chars, optional `0x`). They appear under `boing_getNetworkInfo`.`end_user` so dApps can call `fetchNativeDexIntegrationDefaults` without hardcoding the full router / LP stack (RPC-API-SPEC.md, BOING-DAPP-INTEGRATION.md).

3. RPC Endpoints

Method	Params	Description
`boing_submitTransaction`	`[hex_signed_tx]`	Submit a signed transaction
`boing_chainHeight`	`[]`	Current chain height
`boing_getSyncState`	`[]`	Committed tip: head / finalized / latest block hash
`boing_getNetworkInfo`	`[]`	dApp discovery: tip + timing + optional **`chain_id`** / **`chain_name`** (env); explicit non-availability of chain-wide TVL / APY — RPC-API-SPEC.md
`boing_getBalance`	`[hex_account_id]`	Spendable balance (decimal string)
`boing_getAccount`	`[hex_account_id]`	Balance, nonce, stake (for wallets and tx building)
`boing_getBlockByHeight`	`[height]`	Block at height (u64)
`boing_getBlockByHash`	`[hex_block_hash]`	Block by hash (32 bytes hex)
`boing_getAccountProof`	`[hex_account_id]`	Merkle proof for account
`boing_verifyAccountProof`	`[hex_proof, hex_state_root]`	Verify Merkle proof
`boing_simulateTransaction`	`[hex_signed_tx]`	Simulate tx (gas, success)
`boing_registerDappMetrics`	`[hex_contract, hex_owner]`	Register dApp for incentives
`boing_submitIntent`	`[hex_signed_intent]`	Submit signed intent for solver fulfillment
`boing_faucetRequest`	`[hex_account_id]`	Testnet only: request testnet BOING (node must be started with `--faucet-enable`)
`boing_getLogs`	`[{ fromBlock, toBlock, address?, topics? }]`	Bounded log query (max **128** inclusive heights, **2048** rows per call); see RPC-API-SPEC.md

Public RPC operators and `boing_getLogs`

- **`boing_getLogs` is more expensive than single-block reads** (each call scans up to **128** committed heights and may return up to **2048** log rows). On **shared** or **rate-limited** endpoints, operators should **document** whether the method is enabled and any **extra limits** (e.g. lower QPS than `boing_chainHeight`).

- If you need to protect the node, prefer **global RPC rate limits** (see `RateLimitConfig` / node defaults) or a **reverse proxy** rule; disabling the method is a last resort and breaks indexers that rely on it—see INDEXER-RECEIPT-AND-LOG-INGESTION.md.
- **Indexer clients:** use **`boing-sdk`** **`getLogsChunked`** (or equivalent chunking) so each RPC stays within the **128**-block cap; default **`maxConcurrent: 1`** on wide backfills is polite on public tiers. For full block + receipt replay, **`fetchBlocksWithReceiptsForHeightRange`** and **`planIndexerCatchUp`** are documented in the same indexer guide and in examples/native-boing-tutorial (`npm run indexer-ingest-tick`).
- **Internet-facing smoke (no keys):** from the tutorial package, **`npm run preflight-rpc`** with public **`BOING_RPC_URL`** exercises **`check-testnet-rpc`** plus a one-shot **`getSyncState`** sample; command matrix: PRE-VIBEMINER-NODE-COMMANDS.md.

Example (curl):

```
curl -X POST http://127.0.0.1:8545/ -H "Content-Type: application/json" \
  -d '{"jsonrpc":"2.0","id":1,"method":"boing_chainHeight","params":[]}'
```

4. CLI Usage

Command	Description
`boing init [name]`	Scaffold a new dApp project
`boing dev [--port 8545]`	Start local dev chain
`boing deploy [path]`	Deploy contract or config
`boing metrics register --contract <hex> --owner <hex>`	Register contract for dApp incentives
`boing completions <shell>`	Generate shell completion (bash, zsh, fish, powershell, elvish)

Shell Completion

```
# Bash
boing completions bash > /etc/bash_completion.d/boing  # or ~/.local/share/bash-completion/completions/boin

# Zsh
boing completions zsh > ~/.zsh/completions/_boing

# Fish
boing completions fish > ~/.config/fish/completions/boing.fish
```

5. Monitoring & Health

Chain Height

```
curl -s -X POST http://127.0.0.1:8545/ -H "Content-Type: application/json" \
  -d '{"jsonrpc":"2.0","id":1,"method":"boing_chainHeight","params":[]}' | jq
```

Block Query

```
curl -s -X POST http://127.0.0.1:8545/ -H "Content-Type: application/json" \
  -d '{"jsonrpc":"2.0","id":1,"method":"boing_getBlockByHeight","params":[0]}' | jq
```

Logs

Set `RUST_LOG` before running:

```
RUST_LOG=info cargo run -p boing-node
# Debug: RUST_LOG=debug
# Trace: RUST_LOG=trace
```

6. Incident Response

For security incidents and vulnerabilities:

Step	Action
1. **Detect**	Monitor logs, alerts, community reports.
2. **Assess**	Classify severity: Low, Medium, High, Critical.
3. **Contain**	Isolate affected systems; pause if necessary.
4. **Communicate**	Notify validators, users, ecosystem per severity.
5. **Remediate**	Apply fixes; coordinate upgrades via governance if needed.
6. **Post-mortem**	Document cause, impact, and prevention.

Contacts: SECURITY-STANDARDS.md § **5. Security Contacts** (responsible disclosure via GitHub Security Advisories; bug bounty **TBD**).

6b. Scalability Characteristics

- **Block time:** ~2 seconds (configurable via tokenomics)
- **Throughput:** Parallel transfer batches; access-list batching reduces conflicts
- **Gas:** Fixed per tx type (Transfer, Bond, Unbond, ContractCall, ContractDeploy, QaPoolVote); **fee market v0** charges `ceil(gas_used × GAS_PRICE / GAS_UNITS_PER_BOING)` (1 BOING per 21,000 gas, so a transfer costs 1 BOING) plus optional `extra_fixed_fee` / `transfer_amount_bps`. Default split is **100% protocol treasury** (`PROTOCOL_TREASURY`); block **emission** still goes to the round proposer. Override via `network_fee_config.json` / `boing_operatorApplyFeePolicy` (see TECHNICAL-SPECIFICATION.md §8.4).
- **Batching:** Scheduler groups non-conflicting txs; transfers with disjoint access lists run in parallel

6c. Decentralization Design

- **Permissionless validation:** No whitelist; anyone with stake can validate
- **P2P discovery:** Bootnodes for bootstrap; mDNS for LAN; DHT (roadmap) for discovery
- **No central gatekeeper:** Consensus, governance, and QA pool are decentralized
- **Single-client today:** Multiple implementations encouraged for resilience

7. Troubleshooting

Node won't start

1. Ensure port 8545 (or `--rpc-port`) is free.
2. Check `RUST_LOG=debug` for errors.
3. On Windows: ensure no firewall blocking; try WSL if TCP binding fails.

Transaction not included

- Validator mode must be enabled for block production.
- Check mempool size and nonce ordering.
- Simulate first: `boing_simulateTransaction` to validate.
- **Note:** If block production or consensus fails, transactions are re-inserted into the mempool automatically so they can be retried in the next round.

RPC returns "Method not found"

- Ensure you're using the exact method name (case-sensitive).
- Params must be a JSON array (e.g. `"params": []` not `"params": {}`).

Build fails

```
cargo clean
cargo build
```

8. Testnet Operations

When running the **public incentivized testnet**, the following operations keep bootnodes and the faucet available. See TESTNET.md Part 3 for the full launch checklist.

8.1 Running a bootnode

A **bootnode** is a node with a stable, publicly reachable address that other nodes use to join the network.

1. **Build:** `cargo build --release`
2. **Run with P2P and a fixed port:**

```
./target/release/boing-node \
  --p2p-listen /ip4/0.0.0.0/tcp/4001 \
  --validator \
  --rpc-port 8545 \
  --data-dir ./bootnode-data
```

3. **Publish the multiaddr:** Your bootnode address is `/ip4/<YOUR_PUBLIC_IP>/tcp/4001`. Ensure TCP port 4001 (and 8545 if RPC is public) is open in the firewall. Add this multiaddr to TESTNET.md §6 and to `website/src/config/testnet.ts` (or set `PUBLIC_BOOTNODES` at build time).
4. **Recommendation:** Run at least **two** bootnodes on different hosts for redundancy.
5. **P2P transaction gossip:** After a successful **`boing_submitTransaction`** or **`boing_faucetRequest`**, the node gossips the **signed** transaction on `boing/transactions`. Peers verify the signature and apply the same mempool + QA rules as RPC submissions (duplicates and rejects are dropped at debug log level). **Regression:** `cargo test -p boing-node --test p2p_tx_gossip_rpc` (four-node full mesh: RPC submit on one peer → another peer's mempool contains the same tx id). **Note:** libp2p gossipsub's default mesh size means **two peers alone** may not propagate topic messages reliably; production testnets should run enough connected peers (or tune gossipsub — see `boing-p2p`).
6. **Connections per IP:** The active rate-limit profile caps simultaneous P2P connections from the same remote IPv4/IPv6 address (mainnet profile **50**, dev profile **100**). Override with **`--max-connections-per-ip N`**; use **`0`** for unlimited (not recommended on public listeners).
7. **Multi-validator consensus (optional):** By default the node is **single-validator** (local AccountId `[1u8; 32]`). For a multi-validator testnet, set a shared static set and this node's signing key:

```
cargo run -p boing-node -- \
  --p2p-listen /ip4/0.0.0.0/tcp/4001 \
  --validator \
  --validators <HEX32>,<HEX32>,<HEX32>,<HEX32> \
  --validator-key <HEX32_SECRET> \
  --rpc-port 8545 --data-dir ./data
```

Env equivalents: `BOING_VALIDATORS`, `BOING_VALIDATOR_KEY`. Round-robin leader proposes; followers vote on `boing/votes`; tip advances at **2f+1** quorum. **Regression:** `cargo test -p boing-node --test multi_validator_consensus`.

8. **Stake-derived validator set (optional):** After a static bootstrap set is configured, enable epoch refresh from top stakers:

```
cargo run -p boing-node -- \
  --validator \
  --validator-set stake \
  --stake-validator-top-n 21 \
  --stake-validator-epoch-len 100 \
  --validators <HEX32>,... \
  --validator-key <HEX32_SECRET> \
  --rpc-port 8545 --data-dir ./data
```

Env: `BOING_VALIDATOR_SET=stake`, `BOING_STAKE_VALIDATOR_TOP_N` (default **21**), `BOING_STAKE_VALIDATOR_EPOCH_LEN` (default **100**), `BOING_STAKE_VALIDATOR_MIN_STAKE` (default **10000**). On each commit where `height % epoch_len == 0` (and `height ≥ epoch_len`), the node replaces the consensus set with accounts whose **active stake ≥ min_stake** from `StateStore::top_stakers`, and **always retains the local validator** so this process can still vote. If nobody meets the threshold, the current set is kept. **Unbonding:** `Unbond` queues stake for `UNBONDING_DELAY_BLOCKS` (100); claim with `ClaimUnbond`. **Leader election:** `--leader-election vrf` / `BOING_LEADER_ELECTION=vrf` (default `round_robin`) — ECVRF proofs on `boing/vrf-proofs` when the full set is present, else BLAKE3 stub. **Equivocation slash:** conflicting votes (local or gossiped `EquivocationEvidence`) burn `EQUIVOCATION_SLASH_BPS` (50%) of the offender's active stake to the fee burn sink, record a `SlashRegistry` entry (persisted as `slash_registry.json` when `--data-dir` is set), and allow an in-window appeal (`boing_submitSlashAppeal` / `boing_resolveSlashAppeal`; approved appeals restore stake from the burn sink). **Liveness slash:** non-leaders with pending mempool txs call `tick_liveness`; after `LIVENESS_MISS_THRESHOLD` (3) leader timeouts of `LIVENESS_LEADER_TIMEOUT_SECS`, burn `LIVENESS_SLASH_BPS` (10%). **Regression:** `cargo test -p boing-node --test stake_validator_epochs`; `cargo test -p boing-node --test equivocation_slash`; `cargo test -p boing-node --test liveness_slash`. Remaining gaps (production ECVRF) are tracked in NEXT-STEPS-FUTURE-WORK.md.

8.2 Running the faucet node

The **faucet** is an RPC method (`boing_faucetRequest`) on a node started with `--faucet-enable`. Use a dedicated node (or a node behind your public RPC) so the website faucet page can target it.

1. **Build:** `cargo build --release`
2. **Run with faucet enabled:**

```
./target/release/boing-node \
  --validator \
  --faucet-enable \
  --rpc-port 8545 \
  --data-dir ./faucet-data
```

   For the **public testnet**, also use `--p2p-listen` and `--bootnodes` so this node syncs with the network.
3. **Publish the RPC URL:** Point users and the website to this node's RPC (e.g. `https://testnet-rpc.boing.network/`). Set `PUBLIC_TESTNET_RPC_URL` when building the website so the faucet page defaults to this URL.
4. **Chain metadata on public testnet:** Export **`BOING_CHAIN_ID=6913`** and **`BOING_CHAIN_NAME=Boing Testnet`** before starting **`boing-node`** (or use **`Environment=`** in systemd) so **`boing_getNetworkInfo`** exposes them — `tools/boing-node-public-testnet.env.example`.
5. **Rate limit:** The faucet allows 1 request per 60 seconds per account; no extra config needed.
6. **Genesis:** The faucet account is funded at genesis with 10,000,000 testnet BOING (see `boing-node/src/faucet.rs`). Ensure all testnet nodes use the same genesis so the faucet balance exists.

8.3 Cloudflare Tunnel (testnet-rpc.boing.network)

For full setup steps, see INFRASTRUCTURE-SETUP.md. If you use Cloudflare Tunnel to expose the RPC at `https://testnet-rpc.boing.network/`:

- **"Failed to initialize DNS local resolver"** — This cloudflared log message is usually harmless. The tunnel has already registered; traffic forwarding works. It occurs when cloudflared cannot reach `region1.v2.argotunnel.com` for optional region/metrics. You can ignore it. If it bothers you, try a different DNS (e.g. 1.1.1.1) or firewall rules that allow outbound DNS.

- **HTTP 530 / body `error code: 1033` (or similar) from `https://testnet-rpc.boing.network/`** — Cloudflare is up, but the **tunnel origin** (the machine running `cloudflared` + `boing-node` RPC) is **unreachable** or the tunnel process is down. This is **not** a JSON-RPC or SDK bug: no `boing_qaCheck` /

`boing_chainHeight` will succeed until ops restores the tunnel or you point clients at a **working** RPC (e.g. local `http://127.0.0.1:8545`). The **boing-sdk** surfaces the response body in the error message when the status is non-2xx.

8.4 Monitoring the testnet

- **Upgrading the RPC binary:** PUBLIC-RPC-NODE-UPGRADE-CHECKLIST.md (build, test, tunnel, post-deploy **`check-testnet-rpc`**).
- **Chain height:** Call `boing_chainHeight` on the public RPC periodically; alert if growth stalls. Tutorial **`BOING_POLL_ONCE=1 node scripts/observer-chain-tip-poll.mjs`** emits one JSON sample (see TESTNET-RPC-INFRA.md §3).
- **Faucet balance:** Check the faucet account balance via `boing_getBalance` with the faucet account ID; refill or alert when low (genesis funding is 10M; 1,000 per request).
- **Bootnode reachability:** Ensure ports 4001 (P2P) and 8545 (RPC, if exposed) are reachable from the internet; use a simple TCP check or your monitoring stack.

8.5 Log: `P2P: block publish error: InsufficientPeers` (Gossipsub)

What it means: libp2p Gossipsub only forwards a published message to peers that have **advertised subscription** to the topic (`boing/blocks` / `boing/transactions`). Right after startup—or on very small networks—there can be a **short window** where you are connected (or still dialing) but **no peer is in `topic_peers` yet**, so `publish` returns `InsufficientPeers`.

What is *not* broken: Local consensus already **committed** the block (you will see `Consensus: committed block` in the same trace). JSON-RPC, faucet, and wallet balance use the node's local chain, not Gossipsub.

Propagation: Other nodes can still obtain blocks via the **block request/response** protocol (`/boing/block-sync/1`) once they are connected; Gossipsub is an optimization for fan-out, not the only sync path.

If the second node never catches up: Check bootnode multiaddrs, firewall **TCP 4001**, matching genesis, and that both sides run a build with the same network ID / chain config—not this warning alone.

8.6 “Smart contracts”, boing.finance, and Boing devnet

Boing L1 uses the Boing VM only. Execution is the stack machine + opcodes in `crates/boing-execution`, with bytecode QA in `boing-qa`. Contracts are deployed with on-chain **`ContractDeploy`** / called with **`ContractCall`** payloads inside Boing `Transaction`s, submitted as `boing_submitTransaction` (see `docs/RPC-API-SPEC.md`). **boing.finance** must not treat **chain 6913** as an EVM chain: its `contracts.js` entry uses **32-byte** `AccountId` fields (`nativeConstantProductPool`, optional `nativeVm.*` module ids from build env), not Solidity factory/router addresses.

To get on-chain programs on devnet:

1. **Author bytecode** accepted by the protocol QA gate (see `docs/QUALITY-ASSURANCE-NETWORK.md` and mempool/RPC QA checks).
2. **Build and sign** a `ContractDeploy` transaction with the Boing signing model (BLAKE3 + Ed25519 + bincode), or use tooling that outputs `boing_submitTransaction`-compatible hex (CLI/SDK as they mature).
3. **Run nodes** with the same genesis and peer connectivity so blocks (and deploy txs) propagate.

To extend boing.finance Swap / pools / lockers on Boing: implement **Boing RPC +** `ContractCall` (and Express signing) against operator-published VM contracts; wire their **AccountIds** into env / `contracts.js` `nativeVm`. EVM-only code paths in that app intentionally skip chain **6913**.

Boing Network — Authentic. Decentralized. Optimal. Sustainable.